



Towards Deployable Anomaly Detection in industrial CPS/ICS: A Survey and Deployability Assessment

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Abstract

Anomaly detection is widely used as an early-warning mechanism in CPS/ICS environments, yet detection accuracy alone does not determine whether a detector can be trusted, deployed, and operated under realistic conditions. This survey examines how recent anomaly detection approaches address these practical deployment requirements in CPS/ICS, where detectors face unique operational challenges. Following the PRISMA guidelines, a structured search and multi-stage screening across four academic databases (ACM Digital Library, IEEE Xplore, Scopus, and ScienceDirect) yielded a corpus of 26 peer-reviewed, implementation-based studies. From each study, information on method family, data modality, and datasets was extracted. Furthermore, a set of nine deployability criteria was derived from safety, security, and AI-governance standards and frameworks. Each study was then systematically assessed against these criteria using an evidence-based labeling scheme, with each criterion classified as fully, partially, or not considered based on the evidence reported in the study. The analysis shows that while detection accuracy is consistently reported, evidence for the remaining criteria is often limited or absent. These findings highlight significant gaps between reported performance and deployment readiness.

Keywords Anomaly detection · Safety-critical systems · Cyber-physical systems · Systematic review · Deployability criteria

1 Introduction

Cyber-physical systems (CPS) and industrial control systems (ICS), such as water treatment plants, power grids, and industrial process environments, provide essential services in modern society and are becoming increasingly interconnected and software-dependent. As these systems grow in complexity and connectivity, they become exposed to cyber-

attacks, system failures, sensor degradation, and operational faults that can propagate across both cyber and physical layers [17]. In such environments, delayed or missed detection of abnormal behavior can lead to severe consequences, including physical harm, large-scale service disruption, environmental damage, and significant economic loss [30].

Anomaly detection is widely used as an early warning mechanism to identify deviations from expected operation before they escalate into safety incidents. More specifically, it involves the identification of observations or patterns that deviate from the normal system behavior [8]. In a general pipeline, it consists of the collection of operational data, the modeling of normality through statistical or learning-based representations, and the classification of new observations as normal or anomalous [8]. Depending on the temporal or contextual nature of the data, anomalies may appear as isolated points, collective events, or context-dependent deviations.

Unlike signature-based detection, anomaly detection is particularly suited to CPS/ICS because it can, in principle, identify previously unseen faults and cyberattacks by learning normal system behavior. It can also be applied to diverse data sources, including network traffic, logs, and time-series data collected from sensors and controllers [8]. At the same

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time, the properties that make anomaly detection appealing also make it difficult to validate and deploy in practice [8]. A detector that performs well in offline evaluations may still be unsuitable for real-world deployment if it produces unstable detection results, generates high false alarms, fails under distribution shift, cannot keep pace with streaming data, or introduces excessive computational and memory overhead on resource-constrained edge devices.

In this paper, *deployability* refers to the property of an anomaly detector that is not only accurate in controlled evaluation, but also capable of operating reliably under realistic CPS/ICS conditions. This includes tolerating imperfect or degraded data, meeting timing and resource constraints, remaining effective under changing operating conditions, and providing sufficient assurance for integration into operational workflows. Deployability is therefore treated as a multi-criteria property.

This perspective motivates the central question of this survey: *How do recent anomaly detection approaches address deployability in industrial CPS/ICS?* The challenge is not simply whether a detector can identify anomalies, but whether it can do so under the conditions encountered in real-world deployment. In practice, operators require detectors that remain dependable under degraded sensing, deliver timely and accurate alerts, adapt predictably to evolving operating conditions, and comply with security and governance requirements.

However, existing literature does not consistently evaluate these broader deployment conditions. As a result, practitioners have limited evidence for judging which methods are suitable for operational use, while researchers lack a systematic understanding of progress beyond improvements in detection performance. This motivates the need for a survey that examines not only which anomaly detection methods have been proposed, but also the extent to which they provide evidence for deployability.

To address this need, this survey systematically analyzes recent anomaly-detection approaches for CPS/ICS. A structured search and multi-stage screening process based on the PRISMA guidelines was used to construct an evidence base of 26 implementation-based studies from major academic databases. For each included study, information on method family, data modality, datasets, and reported evidence relevant to deployability was extracted and analyzed. The studies were then assessed against deployability criteria that were derived from safety, security, and AI-governance standards/frameworks.

The main contributions of this paper are:

- A systematic collection and analysis of recent implementation-based anomaly-detection approaches for CPS/ICS.

- A set of deployability criteria for assessing anomaly detection approaches in CPS/ICS.
- An evidence-based assessment of the reviewed studies based on their level of consideration of the defined deployability criteria.
- An overview of key deployability gaps and future research directions.

The remainder of the paper is organized as follows. Section 2 positions this survey relative to existing reviews and explains why a criteria-centric synthesis is needed. Section 3 describes the study selection process, the inclusion/exclusion criteria, and the subsequent methodological steps for criteria definition and assessment. Sections 4–6 analyze the reviewed anomaly detection implementations and assess them against the deployability criteria. Section 7 discusses the implications of the findings and highlights the main gaps observed. Finally, Section 9 concludes the paper and outlines directions for future research.

2 Related Work

This section reviews prior surveys and related work on anomaly detection, focusing on studies that discuss methodological aspects such as taxonomies, assumptions, and evaluation practices relevant to safety-critical systems. Emphasis is placed on studies that provide insights into the design and evaluation of anomaly detection methods under operational conditions rather than purely descriptive application overviews.

A widely cited survey in anomaly detection is [8]. It offers a structured decomposition of anomaly detection problem formulations, including point, contextual, and collective anomalies. It also distinguishes common learning settings including, semi-supervised, and unsupervised approaches, and it discusses typical output types such as anomaly scores or binary labels, alongside major technique families. This survey remains valuable for safety-critical contexts because it connects method families to their underlying assumptions and highlights practical considerations such as computational complexity and the difficulty of obtaining representative anomalous labels, both of which directly affect deployability in safety-critical systems.

Hodge and Austin [18] is an earlier survey, frequently cited in research works for outlier detection. It provides a broad survey of outlier detection methodologies and catego-

size approaches originating from statistics, neural networks, and machine learning, including distance-based, density-based, depth-based, model-based, and graph-based families. Although this work predates much of the modern deep-learning, it remains useful as a positioning reference because it stresses that the choice of an outlier detector is inseparable from assumptions about the data distribution, dimensionality, noise characteristics, and the acceptable trade-off between detection performance and computational cost.

Pimentel et al. [44] conducted a closely related review addressing novelty detection and one-class learning, which often aligns with safety-critical practice where anomalies are rare, evolving, or expensive to label. The authors review novelty detection and synthesize algorithmic families across probabilistic and statistical modeling, reconstruction-based methods, domain-based techniques, information-theoretic approaches, and spectral methods, while also discussing evaluation concerns and operational constraints. This perspective is directly relevant for many safety-critical settings because monitoring pipelines often model normal behavior and treat deviations as novel, rather than assuming a closed-world taxonomy of all fault and attack classes.

Deep learning has become a dominant paradigm for anomaly detection in high-dimensional, multivariate, and weakly labeled settings, and this shift has motivated several dedicated surveys. A central reference in this area is the work by Pang et al. [43], which provides a comprehensive taxonomy of deep learning approaches for anomaly detection and organizes methods by their core modeling strategy and supervision regime, while also discussing the assumptions, strengths, and limitations that distinguish deep anomaly detection from standard deep classification pipelines. For the purpose of this work, its value lies in the explicit articulation of why anomaly detection is challenging for deep learning and in its systematic mapping from these challenges to families of methods such as reconstruction-based detectors, prediction-based detectors, one-class representation learning, and hybrid formulations that incorporate self-supervision or auxiliary objectives.

A complementary perspective is provided by Ruff et al. [46], who present a unifying review that bridges deep and shallow anomaly detection under common modeling dimensions. Their work helps clarify what is genuinely gained by deep representation learning, from what can already be achieved by classical density, distance, or one-class decision rules. It emphasizes that many deep detectors ultimately reduce to classical scoring principles applied in learned feature spaces. This framing is particularly relevant in safety-critical contexts where the added operational cost of deep models must be justified by measurable benefits, such as improved robustness.

Darban et al. [58] provide a dedicated survey of deep learning for time series anomaly detection and explicitly

organize the literature around time-series-specific anomaly detection strategies. In particular, they distinguish major modeling approaches such as forecasting-based detection, reconstruction-based detection, and hybrid designs, and relate these strategies to the deep architectures commonly used in practice. Beyond taxonomy, their survey also compiles commonly used benchmarks and datasets and discusses open challenges for time-series anomaly detection, including issues that are central in safety-critical deployments such as evolving normality and non-stationarity, limited availability of labeled anomalies, and the difficulty of establishing reliable evaluation protocols across heterogeneous application domains.

Some more recent surveys narrow the scope from anomaly detection in general to CPS, motivated by the fact that anomalies in CPS can reflect either cyber intrusions or physical-process failures and can therefore have direct safety implications. Abshari and Sridhar [2] provide a recent survey focused explicitly on anomaly detection in CPS, and categorize approaches into families that include machine learning, deep learning, mathematical, invariant-based, and hybrid techniques. Their survey is useful for positioning because it frames anomaly detection as a CPS reliability and safety mechanism, and it emphasizes practical concerns such as real-time monitoring and adaptation to changing conditions, which are central in operational CPS deployments.

Beyond surveys that focus explicitly on anomaly detection, there are also CPS security surveys that review alternative detection strategies commonly used in safety-critical systems, such as specification-based and invariant-based monitoring. Nweke et al. [41] survey specification-based intrusion detection for CPS and classify the literature by dimensions such as the source of specifications, modeling formalism, detection mechanism, detector placement, and validation strategy. Although this line of work is distinct from data-driven anomaly detection, it is highly relevant in safety-critical contexts because specification- and invariant-based monitors are often preferred when interpretability, auditability, and predictable failure behavior are required. At the same time, specification-based surveys typically do not aim to cover the broader anomaly detection design space, nor do they provide a unified comparison between specification-based monitors and learned anomaly detectors under shared operational constraints.

A further set of surveys focuses on anomaly detection for temporal and streaming data, which is central to safety-critical monitoring because of continuous telemetry, logs, and sensor streams being so common. Gupta et al. [13] survey outlier detection for temporal data and organize the space by temporal data characteristics and modeling strategies, highlighting recurring challenges such as concept drift, evolving normality, and the need for online and incremental processing.

Table 1 Comparison of related literature. ✓: addressed, **P**: partially addressed, -: not addressed

Work	CPS/ICS focus	Method taxonomy	Implementation focus	Deployability dimensions	Evidence-based labeling
Chandola et al. [8]	–	✓	–	P	–
Hodge & Austin [18]	–	✓	–	P	–
Pimentel et al. [44]	–	✓	–	P	–
Pang et al. [43]	–	✓	–	P	–
Ruff et al. [46]	–	✓	–	P	–
Darban et al. [58]	–	✓	–	P	–
Abshari & Sridhar [2]	✓	✓	P	P	–
Nweke et al. [41]	✓	–	–	P	–
Gupta et al. [13]	–	✓	–	P	–
Tambon et al. [52]	–	–	–	✓	–

In addition, a recurring concern in high-stakes deployments is not only detecting anomalies but also controlling the operational cost of alarms. In safety-critical systems, detectors must often be calibrated to achieve predictable false-alarm behavior under changing operating conditions, and they must be evaluated under distribution shift, missingness, and sensor degradation. While these issues are frequently discussed as limitations in anomaly detection surveys [8, 43], they are less often treated as first-class comparative dimensions across methods and domains, which makes it difficult to infer whether reported performance translates to reliable field operation.

A complementary body of work approaches machine learning from the perspective of safety assurance and certification rather than anomaly-detection methodology. The most comprehensive synthesis is the systematic review by Tambon et al. [52], which surveys how machine learning can be certified and organizes the field around six recurring pillars: robustness, uncertainty, explainability, verification, safe reinforcement learning, and direct certification. Several of these pillars map closely onto the deployability criteria used in this survey, which reinforces that these properties are what assurance practice treats as decisive. This assurance perspective is supported in a range of safety and AI-governance standards that define what evidence is expected. Functional-safety standards such as IEC 61508 [23] establish the lifecycle, verification, and residual-risk arguments required in traditional safety-critical engineering. More recent AI-specific instruments, including the NIST AI Risk Management Framework [39] and the EU AI Act [10], extend these expectations toward data quality, robustness, transparency, and human oversight for machine-learning components.

Overall, prior surveys provide strong foundations on anomaly detection taxonomies [8, 18, 44], deep anomaly detection methods [43, 46], CPS-oriented anomaly detection [2], specification-based monitoring for CPS security [41], and the certification of machine-learning sys-

tems [52]. Table 1 shows, however, that no existing review combines an implementation-oriented corpus, an explicit set of deployability dimensions, and evidence-based labeling of individual studies. This gap is important because evidence of deployability is often reported inconsistently or absent, making it difficult for practitioners to determine which methods are suitable for real-world deployment.

3 Methodology

This study systematically analyzes recent anomaly-detection approaches for industrial CPS/ICS and evaluates the extent to which they address deployability. Specifically, it (i) identifies recent anomaly-detection studies that include implementations from major academic databases, (ii) defines deployability criteria, (iii) assesses each included study based on its level of consideration of these criteria, and (iv) synthesizes the key deployability gaps. These objectives are guided by the central research question: *How do recent anomaly detection approaches address deployability in industrial CPS/ICS?* Figure 1 provides an overview of the methodological pipeline, showing the study selection process, the deployability criteria derivation from safety, security, and AI-governance standards/frameworks, and the per-study assessment that reveals gaps and opportunities.

3.1 Study selection and screening

To construct a transparent evidence base, the study selection and screening followed the PRISMA guidelines, which standardize the reporting of search strategies, screening steps, and inclusion/exclusion decisions [35, 42]. This approach makes the identification of the selected studies explicit and traceable, reducing selection bias and improving reproducibility.

Four primary academic databases were used to collect relevant scientific works: ACM Digital Library, IEEE Xplore,

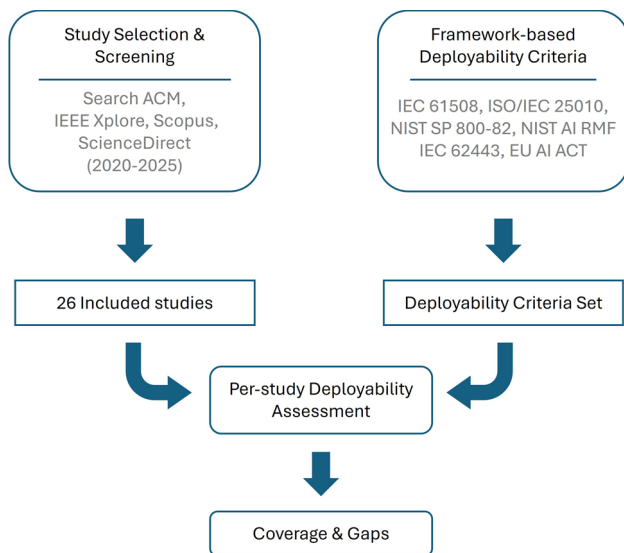


Fig. 1 Overview of the methodology

Scopus, and ScienceDirect. These sources were selected to cover peer-reviewed journal and conference publications.

The search covered publications from January 2020 to October 2025 to capture recent implementation-based anomaly detection approaches. A unified Boolean query was defined and adapted to each platform's syntax to ensure consistency across databases. In each database, the query was applied to titles to ensure that anomaly detection in CPS/ICS settings is the central topic of the collected records.

The query combined two parts: (i) anomaly-detection synonyms (anomaly detection, outlier detection, fault detection) and (ii) CPS/ICS domains (CPS, cyber physical systems, industrial control, ICS, SCADA, power system, smart grid, water treatment, water distribution), using an AND between the two groups and OR within each group.

The database searches yielded a total of 452 records, distributed as follows: 28 from ACM, 84 from IEEE Xplore, 47 from ScienceDirect, and 293 from Scopus. After the removal of 47 duplicate entries, 405 unique records remained for screening.

Figure 2 summarizes the study identification and screening process, showing the number of works retained in each stage. To ensure relevance, technical depth, and comparability, we applied the inclusion and exclusion criteria of Table 2 sequentially during title, abstract, and full-text screening. Title-level screening removed 313 records, abstract screening excluded a further 43 works, and full-text screening removed 23 more. This procedure yielded a final industrial CPS/ICS implementation corpus of 26 studies. The full-text screening

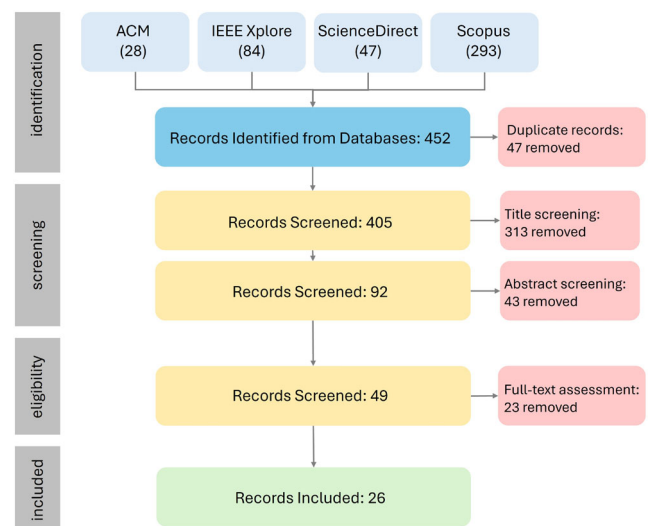


Fig. 2 PRISMA flow diagram for the study selection and screening process

log, together with the inclusion/exclusion codes assigned to each record, is available in a [GitHub repository](#)¹.

Table 2 summarizes the inclusion and exclusion criteria used throughout the study. In cases where multiple publications described substantially overlapping methods, the most recent and methodologically detailed peer-reviewed version was retained to avoid over-representing a single approach family and to maintain diversity in the final set.

To reduce subjectivity during synthesis and support the criteria-based assessment, a structured data-extraction protocol was applied to all included studies. For each paper, we extracted the following information:

1. the anomaly-detection approach.
2. the data modality (e.g., multivariate time series, network traffic).
3. the datasets used.

This information was used to categorize the studies into the four methodological families and to provide a concise description of each approach in Section 4.

3.2 Deployability criteria derivation

To assess how deployability is considered by each anomaly detector, we define a set of criteria grounded in safety, security, and software quality standards and frameworks. In particular, the criteria are derived from IEC 61508 [23], ISO/IEC 25010:2023 [25], NIST SP 800-82 [50], IEC 62443

¹ GitHub repository: <https://github.com/nkollarou/Towards-Deployable-Anomaly-Detection-in-industrial-CPS-ICS>

Table 2 Inclusion and exclusion criteria used in the literature screening

Inclusion	
Code	Description
IC1	Published between 2020 and 2025.
IC2	Peer-reviewed journal article or conference paper.
IC3	Presents a concrete anomaly detection implementation.
IC4	Evaluated in an ICS/SCADA/CPS/water/power domain or other physical/industrial OT context.
IC5	Reports quantitative evaluation metrics.
IC6	Evaluated on a recognised basis such as (a) named public testbeds, (b) named power-system models, or (c) clearly described real-world operational data.
Exclusion	
Code	Description
EC1	Survey, review, mapping, meta-analysis, or bibliometric studies; book, book chapter, or preprint.
EC2	Book, book chapter, or preprint with no peer-reviewed version.
EC3	Out-of-scope domains, such as automotive/ADAS, and medical/healthcare.
EC4	Publication year earlier than 2020.
EC5	Evaluated exclusively on generic IT/OT datasets with no ICS-specific testbed or data.
EC6	Purely theoretical or proposals with no detector implementation and no empirical evaluation.
EC7	Duplicates/superseded versions: Methodologically redundant or de-duplicated studies, excluded when the detector's algorithm family and any special capability it offers are represented by other retained papers.

[24], NIST AI Risk Management Framework [39], and the EU AI Act [10].

Section 5 presents the criteria derived from the standards using the following process.

- Attribute extraction: Extract attributes from each standard/framework that are applicable to and observable within anomaly-detection approaches.
- Attribute clustering: Cluster synonymous or closely related attributes across the standards/frameworks into candidate deployability criteria.
- Retention rule: Retain a candidate criterion only if it is supported by at least two standards/frameworks, is assessable from published studies, and represents a conceptually distinct property.

The resulting criteria cover detection accuracy, operational resilience, adaptability, real-time responsiveness, resource efficiency, adversarial robustness, data confidentiality, explainability, and verification and assurance. The detailed definitions of these criteria and their derivation process are presented in Section 5.

Each included study was then assessed against each criterion and assigned a level of consideration based on its reported design objectives, implementation, and experimental evidence. For a given criterion, a study was labeled as *fully considered* when the criterion was explicitly identified as a design objective and supported by implementation or experimental evaluation. It was labeled as *partially considered* if the criterion was mentioned or implied by a design choice but was neither explicitly targeted or evaluated. Otherwise, it was labeled as *not considered* when no relevant discussion or supporting evidence was reported.

The assessment was conducted independently by two assessors. Inter-rater reliability was substantial, with Cohen's $\kappa = 0.78$ and an observed agreement of 86%, indicating a high level of consistency beyond chance agreement [9, 31]. Disagreements were resolved through discussion and consensus based on the full text of the corresponding study. The assessment result is described in Section 6, which forms the basis for the synthesis of gaps and opportunities discussed in Section 7.

4 Anomaly Detection Approaches

The field of anomaly detection encompasses a wide range of approaches that balance different factors, such as interpretability, data requirements, and computational cost, against detection performance capability. In industrial CPS/ICS settings, additional factors need to be taken into consideration to determine whether a detector can be trusted and operated in practice.

The literature collected in this survey spans a number of application sectors, reflecting the diversity of CPS/ICS and their data types. To provide a coherent synthesis across these different contexts, the 26 selected studies are categorized into four methodological families and described in subsections below.

- i. **Mathematical approaches**, which derive decision rules from first principles (e.g., residual tests, filtering, change-point detection, and extreme-value modeling).
- ii. **Machine learning approaches**, which learn discriminative structure from engineered or extracted features.
- iii. **Deep learning approaches**, which learn representations directly from raw data.

- iv. **Hybrid approaches**, which combine rule-based or model-based components with learned modules to balance coverage, controllability, and safety-critical criteria.

4.1 Mathematical Approaches

The mathematical family includes approaches that derive decision rules from fundamental principles. Table 3 summarizes the mathematical approaches reviewed.

Zhu et al. [62] explore anomaly detection for ICS and propose an unsupervised invariant-rule mining method based on association rule mining. This method aims to improve detection in settings where labeled attacks are scarce and residual-based models can miss stealthy manipulations. The authors' main contribution is a generation strategy that segments sensor time series and encodes temporal features into predicates. This enables the mining of invariant rules that capture temporal dependencies among physical variables rather than only instantaneous states. Experimental results on the SWaT and WADI benchmarks show good detection performance over other baselines. Furthermore, low latency and early-warning capability are reported in attack scenarios. The study concludes that temporal feature-aware invariant mining yields a portable detector for ICS physical process monitoring. In parallel, limitations are reported, including higher false positives due to cross-segment predicate dependency and reduced sensitivity, motivating future refinements.

Zhang et al. [60] investigated real-time anomaly detection for CPS and proposed a dual-path method called PLS-TWE. This method combines a penalty least-squares detector with a time-window entropy monitor to improve generalization and reduce false positives in non-stationary conditions. The authors' main contribution is the integration of tuned least-squares penalties, which stabilize residuals, with an information-theoretic entropy test that filters suspicious spikes, and a simple localization heuristic that ranks likely fault sources. The results on benchmarks show improvements compared to baselines, leading to the conclusion that a residual plus entropy design provides interpretable and deployable detection.

Jin et al. [27] explore anomaly detection for forecasting-supported power-system state estimation. They propose a framework that uses the innovation-reduction properties of an iterated extended Kalman filter (IEKF) together with the normalized residual of a static state estimator to detect and distinguish sudden load changes, problematic data, and topology errors. The authors' main contribution is the use of an innovation-reduction factor (Inred) and residual decision logic with statistically selected thresholds to separate these anomaly classes. Simulation results on IEEE 14-, 39-, 57-, and 118-bus systems show that the method achieves high detection and discrimination accuracy. In parallel, it remains robust under varying measurement noise, measure-

Table 3 Summary of reviewed mathematical approaches

Reference	Year	Data type	Dataset(s)
Zhu et al. [62]	2025	Multivariate time series	– SWaT [12] – WADI [4]
Zhang et al. [60]	2023	Multivariate time series; network traffic	– SWaT [12] – ICIDS2017 [48]
Jin et al. [27]	2022	Multivariate time series	– Simulated data on IEEE 14-, 39-, 57-, and 118-bus systems
Hosseinzadehtaher et al. [19]	2020	Multivariate time series	– Historical load profile and temperature data

ment redundancy, and network size. It is concluded that combining IEKF innovation reduction with static state estimator residual provides a practical anomaly detector for operational power grids.

Hosseinzadehtaher et al. [19] explore anomaly detection for smart distribution power systems. They propose a proactive intrusion detection system (IDS) targeting false-data manipulation of AMI readings and smart-inverter measurements. The authors' main contribution is a condition monitoring vector (CMV) that compares instantaneous aggregated field measurements against a learned ultra-short-term demand forecasting (USTDF) estimate derived from historical load and temperature. Anomalies are flagged when the CMV violates predefined upper/lower error bands. A decision-making module is integrated to account for intentional control-center actions versus cyber events. Case-study-based evaluations show that accurate ultra-short-term forecasting can tighten the expected operating band and enable timely detection of stealthy false-data injection patterns. The study concludes that forecasting-aided CMV thresholding can strengthen grid-edge cybersecurity in distribution networks, with effectiveness depending on forecasting fidelity and careful calibration of tolerance bands to balance missed detections and false alarms.

4.2 Machine Learning Approaches

The machine-learning family covers detectors that learn discriminative structure from engineered or extracted features. Table 4 summarizes the reviewed machine-learning approaches.

Birihanu and Lendák [7] propose an explainable, correlation-based anomaly detection method for ICS. Their work is motivated by the complex temporal correlations among many ICS devices and the limited interpretability sup-

Table 4 Summary of reviewed machine learning approaches

Reference	Year	Data type	Dataset(s)
Birihanu and Lendák [7]	2025	Multivariate time series	– SWaT [12] – HAI [49] – IoT Modbus dataset
Mokhtari et al. [36]	2021	Measurement data (SCADA)	– HAI [49]

port of existing detectors. The main contribution is a detector that models temporal correlations among parameters, selects informative parameters, and provides interpretable attributions and root-cause identification while operating efficiently with limited resources. Evaluated on ICS benchmarks, the method reports competitive detection together with explanations that indicate which correlated parameters drive an alarm.

Mokhtari et al. [36] propose a machine-learning approach to attack detection in ICS that operates on measurement data rather than network traffic, motivated by the observation that network-based intrusion detection can be deceived by attackers who imitate the system's normal activity. The authors' main contribution is a Measurement Intrusion Detection System (MIDS) that monitors physical measurement data in the SCADA layer, enabling the detection of abnormal activity even when an attacker conceals it at the control layer. A supervised classifier is trained to separate normal from abnormal operation. The reported results show strong detection performance, particularly for stealthy attacks, with a Random Forest classifier outperforming the other tested models on the measurement data. The study concludes that measurement-based detection complements network IDS for ICS, while its reliance on labeled testbed data and supervised training is a limitation.

4.3 Deep Learning Approaches

The deep-learning family comprises methods that learn representations directly from raw data such as multivariate time series or network traces. Table 5 summarizes the reviewed deep-learning approaches.

Liu et al. [33] propose TFANet, a time–frequency fusion feature-attention encoding network for anomaly detection in industrial control system operation data, motivated by the complex, multi-dimensional, long-sequence nature of ICS time series with varying periodicity and misaligned temporal associations. The main contribution is a time–frequency fusion module with attention that captures inconsistent feature periodicity and long-range dependencies, together with the construction of a dataset (OTS) containing thirteen attack types. Evaluated on SWaT, WADI, and OTS,

TFANet reports improved detection metrics over data-driven baselines. The authors conclude that time–frequency-aware attention strengthens ICS operation-data monitoring, while latency and resource behavior are not assessed.

Yang et al. [57] pose the problem of detecting anomalies in CPS time series as an online threshold control problem. They propose ADT, which treats the selection of thresholds over autoencoder-generated anomaly scores as a Markov decision process, solved using a Deep Q-Network (DQN). The agent observes recent scores and contextual cues from a simulated CPS environment and learns actions that balance true/false decisions through a reward shaped for detection quality. Evaluated on SWaT, WADI, and HAI, ADT consistently surpasses fixed or heuristic thresholds, reporting near-perfect F1 scores and retaining robustness when feedback is noisy, delayed, or partial. The study concludes that deep reinforcement learning offers a lightweight, adaptive controller that can sit atop existing detectors to improve decision-time performance, with future work aimed at broader real-system validation and embedded compute/latency analysis.

Zare et al. [59] propose a real-time network-based anomaly detection system for ICS that targets data-manipulation attacks on Modbus/TCP SCADA traffic, which can mislead the control center and alter field commands. The main contribution is an LSTM-autoencoder trained on normal traffic with a precision–recall-curve threshold, designed to process packet sequences online and to detect subtle manipulation and man-in-the-middle attacks. Evaluated on realistic SCADA traffic, the method reports per-attack recall gains over two existing network anomaly detectors. The authors conclude that an online LSTM-autoencoder is well suited to real-time SCADA traffic monitoring, while drift handling and interpretability of decisions are not addressed.

Liao et al. [32] propose DPDGAD, a dual-process dynamic graph-based detector for multivariate time-series anomaly detection in CPS. The main contribution is a dual-process design that couples a dynamic graph-construction branch, which learns evolving inter-sensor relationships, with a temporal branch built on a variational autoencoder and LSTM. This is coupled with an anomaly-explanation analysis that attributes detections to contributing variables. DPDGAD reports improved detection over baselines while offering interpretable attributions. The authors conclude that jointly modeling dynamic graph structure and temporal dependencies improves multivariate detection for CPS, with calibrated confidence and assurance left for future work.

Aslam et al. [6] propose an improved autoencoder approach (ImpAE) for anomaly detection in ICS, motivated by the high false-positive rate and computational complexity of prior single-model detectors. The main contribution is an autoencoder design that targets reduced false positives and lower computational cost while maintaining detection accuracy, with sensor- and actuator-level indications of flagged

points. ImpAE reports improved precision and accuracy relative to baseline autoencoders at reduced complexity. The authors conclude that a refined autoencoder offers an accurate and more efficient detector for ICS, while drift, adversarial robustness, and privacy remain outside scope.

Tang et al. [53] propose GRN, an interpretable multi-variate time-series anomaly detection method for ICS that combines neural graph networks with gated recurrent units, motivated by the need to balance detection performance, interpretability, and training efficiency. The main contribution is a model that automatically learns potential correlations between sensors and uses them to produce both accurate detections and reliable interpretations of which relationships drive an alarm. Evaluated on SWaT and WADI, GRN reports competitive detection together with interpretable outputs. The authors conclude that graph-based GRU modelling yields an interpretable and efficient detector for ICS, while drift handling and resource behavior are not explicitly evaluated.

Macas et al. [34] present an attention-based deep generative model for anomaly detection in ICS. The main contribution is a variational-autoencoder architecture with convolutional encoder and decoder that extracts spatial and temporal features, augmented by an attention mechanism that focuses on the most informative signal regions. Evaluated on the SWaT testbed with the full suite of detection metrics, the model reports strong detection and is additionally probed under input perturbation. The authors conclude that attention-augmented generative modeling improves ICS anomaly detection, while a formal attacker model, uncertainty, and timing are not assessed.

Kea et al. [28] enhance anomaly detection in distributed power systems using an autoencoder-based federated learning approach (FLAE). The main contribution is a federated autoencoder that is trained collaboratively across local nodes, keeping raw user and device data local while reducing central data transfer and response-time delay. Evaluated on public power-consumption data, the method reports improved detection together with reduced communication overhead. The authors conclude that federated autoencoding offers privacy-preserving, communication-efficient detection for distributed power systems, with concept drift and adversarial robustness of the federated scheme left for future study.

Yang et al. [56] propose an unsupervised traffic anomaly detection pipeline for industrial CPS that improves performance of payload-based detection when labels are limited, and traffic exhibits both long-term and short-term temporal dependencies. Their approach starts with a preprocessing stage and then leverages an unsupervised word-segmentation model that uses an LSTM to maintain the semantic structure of payloads. It subsequently employs an unsupervised classifier that integrates a 1D-CNN with BERT-based representations within an autoencoder-style framework, enabling

the model to capture both local and global patterns in the traffic. Their findings indicate that accurate payload segmentation combined with joint modeling of both long-term and short-term dependencies improves industrial traffic anomaly detection without labeled data.

Sahin et al. [47] propose a CNN-LSTM framework for unsupervised anomaly detection in water-distribution plants. The main contribution is a combined convolutional and recurrent network that extracts spatial features with the CNN and temporal dependencies with the LSTM, preceded by linear interpolation to handle missing measurements. The method reports competitive precision, recall, and accuracy. The authors conclude that combining convolutional and recurrent representations is effective for unsupervised water-system monitoring, while adversarial robustness, uncertainty, and resource use are not addressed.

Do Thu Ha et al. [14] propose an explainable anomaly detection method for industrial control system cybersecurity. The main contribution couples an LSTM-autoencoder detector with gradient-based SHAP explanations, so that detected anomalies can be attributed to the contributing measurements. Evaluated on ICS data the method reports effective detection together with feature-level explanations of its decisions. The authors conclude that integrating explainability into an LSTM-autoencoder improves the actionability of ICS alarms, while uncertainty, resilience, and resource behaviour are not evaluated.

Zhou et al. [61] propose FSL-SCNN, a few-shot learning model based on a Siamese convolutional neural network, to address anomaly detection in industrial cyber-physical systems where labeled attack data are scarce. The main contribution is a Siamese architecture with an intelligent optimization of its loss components that generalizes to novel attack classes from few labeled samples, targeting timely detection in large CPS data streams. Reported results show improved detection with limited labels relative to comparison methods. The authors conclude that few-shot learning mitigates label scarcity for industrial CPS security, while concept drift over time and interpretability of the learned similarity remain open.

Huong et al. [22] propose a federated-learning approach for detecting cyberattacks in IIoT-based smart manufacturing, motivated by the need to detect attacks at the edge without centralizing sensitive operational data. The main contribution is an edge-deployed detection architecture trained via federated learning, with a detection threshold that targets contextual noise and concept drift and updates as normal behavior evolves. Evaluated on industrial datasets, the method reports effective detection while reducing bandwidth and transmission time relative to centralized deep learning. The authors conclude that edge-based federated detection provides resilient, privacy-preserving, and timely monitoring

Table 5 Summary of reviewed deep learning approaches

Reference	Year	Data type	Dataset(s)
Liu et al. [33]	2024	Multivariate time series	– SWaT [12] – WADI [4] – Oil-transportation simulation
Yang et al. [57]	2024	Multivariate time series	– SWaT [12] – WADI [4] – HAI [49]
Zare et al. [59]	2024	Network traffic sequences	– SWaT [12]
Liao et al. [32]	2024	Multivariate time series	– SWaT [12] – WADI [4] – BATADAL [3] – SMD [51] – PSM [1] – SMAP [21] – Energy-storage dataset
Aslam et al. [6]	2024	Multivariate time series	– SWaT [12]
Tang et al. [53]	2023	Multivariate time series	– SWaT [12] – WADI [4]
Macas et al. [34]	2023	Multivariate time series	– SWaT [12]
Kea et al. [28]	2023	Multivariate time series	– Electric power consumption [16]
Yang et al. [56]	2023	Network traffic (payload)	– CPAD
Sahin et al. [47]	2023	Multivariate time series	– WADI [4]
Do Thu Ha et al. [14]	2022	Multivariate time series; network traffic	– Gas pipeline dataset [37]
Zhou et al. [61]	2021	Network traffic features	– UNSW-NB15 [38]
Huong et al. [22]	2021	Multivariate time series	– Gas pipeline (SCADA) [54] – ECG [29] – Respiration [29] – Power demand [29] – Gesture [29] – Space shuttle [29] – NYC taxi [40]

for industrial CPS, with robustness to attacks on the learning process identified as future work.

4.4 Hybrid Approaches

Hybrid approaches combine model-based or rule-based components with learned modules to balance trade-offs. Table 6 summarizes the reviewed hybrid approaches.

Jeffrey et al. [26] investigate anomaly detection in heterogeneous CPS environments with scarce labels by proposing a hybrid pipeline that combines domain driven checks with behavioral machine learning. Signature based and threshold-based modules handle obvious or rule-captured events, while an ensemble of machine-learning classifiers addresses more complex multivariate cases, with a divide-and-conquer router and score-fusion layer leveraging operational technology knowledge to reduce training requirements and minimize false alarms. The approach achieves higher accuracy than standalone machine learning baselines and improves robustness across plants by exploiting the complementary error profiles of the different detectors. The study demonstrates

that integrating lightweight, interpretable rule-based stages with ensemble machine learning offers a practical balance of accuracy, efficiency, and transferability for CPS anomaly detection, highlighting standardizing fusion strategies and enabling online adaptation under non-stationary conditions as promising directions for future work.

Hu et al. [20] propose a hybrid outlier detection and state-correction strategy for smart grids that combines an extended Kalman filter with data compensation, motivated by the vulnerability of state estimation to false-data-injection attacks and sudden load changes. The main contribution is a forecast-assisted, EKF-based detector that generates predicted estimates to flag outliers and then applies data compensation to correct the state, so that detection and recovery are addressed jointly. Evaluated on smart-grid (IEEE bus) test cases, the method reports accurate detection of false-data injection and sudden load changes together with improved state estimation under attack. The authors conclude that coupling model-based filtering with data compensation provides a resilient detect-and-correct mechanism for grid state estimation, distinguishing it from detection-only approaches.

Hao et al. [15] explore real-time anomaly detection in industrial CPS/ICS with pronounced seasonality and non-linear drift by coupling a seasonal ARIMA (SARIMA) forecaster with an LSTM that models residual dynamics. SARIMA captures periodic structure to produce a baseline prediction, the LSTM learns remaining nonlinear behaviors, and the combined prediction errors are converted into anomaly scores with a low-latency online thresholding scheme. Findings show that this hybrid pipeline raises accuracy and F1 while cutting false alarms compared with single-model baselines, yet remains deployable under real-time constraints. The authors conclude that decomposing time-series structure into statistical seasonality plus learned nonlinear residues provides a stable, precise detector for operational CPS.

Asefi et al. [5] address anomaly detection and classification in power-system state estimation, focusing on distinguishing bad data, sudden load changes, and false data injection attacks (FDIA). They propose a two-stage hybrid approach. First, an analytical detection layer combines conventional chi-square testing (for bad data in WLS static state estimation) with an anomaly detection index computed from discrepancies between WLS SSE and EKF-based forecasting-aided state estimation. Second, supervised ML is used to classify the detected anomaly as SLC or FDIA and to localize its origin. MRMR feature selection is applied to reduce training complexity. To improve robustness under topology changes, they train the classifier using bus-related features, aiming to avoid retraining when the network configuration changes but bus count remains constant, and demonstrate effectiveness on simulated scenarios using the IEEE 14-bus test system. Overall, the study shows that combining model-based detection signals with data-driven classification can both detect stealthy anomalies and reliably separate operational disturbances from cyberattacks.

Wu et al. [55] present PGRGAT, an end-to-end physics-informed gated recurrent graph attention network for unsupervised anomaly detection in industrial cyber-physical systems, motivated by the need to combine prior process knowledge with data-driven modelling. The main contribution is a physics-informed graph-structure learning module that injects domain knowledge into a gated recurrent graph attention model, improving both modelling performance and interpretability over purely data-driven methods. Evaluated on industrial CPS benchmarks, PGRGAT reports improved detection while offering physically meaningful structure. The authors conclude that physics-informed graph learning yields an accurate and more interpretable detector for ICPS, with deployment-oriented evaluation as future work.

Raman and Mathur [45] propose PbNN, a hybrid physics-based, data-driven framework for detecting cyber-physical attacks through the anomalies they induce in ICS process dynamics. The main contribution is the use of ICS design

knowledge to define the relationships among correlated components, which are then modelled from operational data with a deep convolutional network, yielding a detector grounded in both physics and data. The framework is implemented and evaluated on an operational secure water-treatment plant subjected to real-time stealthy and coordinated attacks, where it reports effective detection. The authors conclude that embedding process design knowledge into a data-driven model improves detection of stealthy process attacks, with the residual-based design offering interpretable indications of affected variables.

Feng and Tian [11] propose Neural System Identification and Bayesian Filtering (NSIBF), which learns a neural state-space model through an encoder–transition decoder. Then it performs density-based detection by applying Bayesian filtering to track hidden-state uncertainty and score observations using a Mahalanobis-distance likelihood criterion. The method is evaluated on a synthetic noisy CPS and on three real CPS benchmarks (PUMP, WADI, SWAT), where NSIBF outperforms multiple baselines and its own residual-only variants. Overall, the work frames anomaly detection as neural system identification plus probabilistic state estimation, showing that explicitly modeling uncertainty via Bayesian filtering yields more robust CPS anomaly detection than pure deep residual approaches.

5 Deployability Criteria for Anomaly Detectors

An anomaly detector is a component of an operational safety function, rather than a stand-alone classifier. Therefore, detection performance alone is insufficient to determine its deployability. A detector must also be trusted, integrated, and maintained under real-world operating conditions, where sensing is imperfect, operating conditions evolve, and timing and resource constraints must be satisfied.

To assess the deployability of anomaly-detection approaches, we first clarify the properties that should be considered during their design, evaluation, and deployment. We define deployability criteria as the properties that an anomaly detector should consider when intended to operate in CPS/ICS environments.

Following the methodology outlined in Section 3, we derive a set of deployability criteria from established safety, security, and software quality standards and frameworks, including IEC 61508 [23], ISO/IEC 25010:2023 [25], NIST SP 800-82 [50], IEC 62443 [24], NIST AI Risk Management Framework [39], and the EU AI Act [10]. We first extract attributes that (i) are applicable to anomaly-detection approaches and (ii) can be assessed from the reported detector design and evaluation. Related attributes are then clustered into candidate deployability criteria, and the following three

Table 6 Summary of reviewed hybrid approaches

Reference	Year	Data type	Hybridization type	Dataset(s)
Jeffrey et al. [26]	2024	Network traffic statistics	Rule/signature-based and ML ensemble	– Generated from a scaled-down commercial greenhouse facility testbed.
Hu et al. [20]	2024	Power-system measurements	Model-based filtering (EKF) and data-driven	– Simulated IEEE 14-, 30-, and 118-bus systems (Matpower).
Hao et al. [15]	2023	Network traffic statistics	Statistical (SARIMA) and machine learning	– Testbed built from power generation, gas pipeline, and railway transportation systems.
Asefi et al. [5]	2023	Multivariate time series	Model-based state estimation and ML classifier	– Simulated dataset on the IEEE 14-bus test system.
Wu et al. [55]	2023	Multivariate time series	Physics-informed and deep learning (graph attention)	– SWaT [12] – WADI [4]
Raman and Mathur [45]	2022	Multivariate time series	Physics-based and deep learning (CNN)	– SWaT (operational secure water-treatment plant) [12] – WADI [4] – BATADAL [3]
Feng and Tian [11]	2021	Multivariate time series	Neural state-space and Bayesian filtering	– SWaT [12] – WADI [4] – PUMP: dataset collected from a water pump system of a small town

retention rules are applied to decide which candidates are retained as final deployability criteria:

- **R1 – Multi-framework support:** The candidate is supported by at least two of the considered standards or frameworks.
- **R2 – Assessability:** The candidate can be assessed from reported detector evidence, such as design choices or evaluation metrics.
- **R3 – Conceptual distinctness:** The candidate represents a conceptually distinct property and is not subsumed by another retained criterion.

Table 7 presents the candidate deployability criteria, their clustered attributes, the supporting standards and frameworks, and the outcome of the retention process (R1–R3), indicating whether each candidate criterion was retained as a final deployability criterion or discarded. The nine final deployability criteria are described below.

1. **Detection Accuracy (ACC)** Detection accuracy captures how well an anomaly detector distinguishes normal from abnormal behavior, including the trade-off between missed detections (false negatives) and false alarms (false positives). It is commonly quantified using standard performance metrics, such as precision, recall, F1-score, and related measures derived from the confusion matrix.

2. **Operational Resilience (OR)** Operational resilience refers to the ability of an anomaly detector to continue operating reliably under non-ideal, but non-malicious conditions. These conditions include sensor noise, missing or delayed measurements, component faults, or other disturbances. OR reflects the detector's ability to maintain stable and dependable detection despite the imperfect and uncertain data encountered in real CPS/ICS environments, without degrading significantly or behaving unpredictably.
3. **Adaptability (AD)** Adaptability refers to an anomaly detector's ability to accommodate gradual changes in normal system behavior over time due to equipment aging, maintenance activities, seasonal demand, or evolving operating conditions. An adaptable detector should continue to detect anomalies accurately despite these changes without requiring frequent retraining or manual reconfiguration.
4. **Real-Time Responsiveness (RTR)** Real-time responsiveness refers to the capability of an anomaly detection system to identify anomalies and produce detection results within the strict time constraints of the target application. In critical CPS/ICS environments, an accurate detection may have limited value if it arrives too late to enable a timely response. Therefore, this criterion assesses whether a detector can satisfy the latency and

Table 7 Derivation of deployability criteria: candidate criteria with their clustered attributes, supporting frameworks, and retention decision

Candidate criterion	Clustered attributes	Supporting frameworks	Decision
Detection Accuracy	Diagnostic coverage; fault detection; functional correctness/completeness/appropriateness; continuous monitoring; anomaly detection; system integrity (FR3)	IEC 61508 ISO/IEC 25010 NIST SP 800-82 IEC 62443 NIST AI RMF EU AI Act	Retain
Operational Resilience	Hardware fault tolerance; safe failure fraction; common cause failure; reliability (availability, fault tolerance, recoverability); fail safe; resilience; recovery; resource availability (FR7)	IEC 61508 ISO/IEC 25010 NIST SP 800-82 IEC 62443 NIST AI RMF EU AI Act	Retain
Adaptability	Management of change; adaptability; modifiability; configuration/change management	IEC 61508 ISO/IEC 25010 NIST SP 800-82 NIST AI RMF EU AI Act	Retain
Real-Time Responsiveness	Fault-tolerant time interval; time behaviour; real-time / time-critical requirements; timely response to events (FR6)	IEC 61508 ISO/IEC 25010 NIST SP 800-82 IEC 62443	Retain
Resource Efficiency	Resource utilization; capacity; scalability; resource constraints / limited computing resources; resource availability (FR7)	ISO/IEC 25010 NIST SP 800-82 IEC 62443	Retain
Adversarial Robustness & Integrity	Integrity; authenticity; non-repudiation; resistance; access control; identification & authentication control (FR1); use control (FR2); system integrity (FR3); security levels (SL1–4)	ISO/IEC 25010 NIST SP 800-82 IEC 62443 NIST AI RMF EU AI Act	Retain
Data Confidentiality	Confidentiality; confidentiality (CIA); data confidentiality (FR4); restricted data flow (FR5)	ISO/IEC 25010 NIST SP 800-82 IEC 62443 NIST AI RMF EU AI Act	Retain
Explainability	Fault annunciation; analysability; accountability; hazard warning; audit and accountability; incident response; audit records (FR2)	IEC 61508 ISO/IEC 25010 NIST SP 800-82 IEC 62443 NIST AI RMF EU AI Act	Retain
Verification	Verification; validation; functional safety assessment; systematic capability; SIL; testability; security assessment; secure development lifecycle	IEC 61508 ISO/IEC 25010 NIST SP 800-82 IEC 62443 NIST AI RMF EU AI Act	Retain
Uncertainty Quantification	Probability of dangerous failure (PFD/PFH)	IEC 61508	Drop (R1)
Risk Identification / Management	Risk identification; risk management	ISO/IEC 25010 NIST AI RMF	Drop (R2)
Usability / Interaction	Interaction capability (learnability, operability, etc.)	ISO/IEC 25010 EU AI Act	Drop (R3)

Table 7 continued

Candidate criterion	Clustered attributes	Supporting frameworks	Decision
Compatibility / Interoperability	Co-existence; interoperability	ISO/IEC 25010	Drop (R1)
Portability / Deployment	Installability; replaceability	ISO/IEC 25010	Drop (R1)
Modularity / Reusability	Modularity; reusability	ISO/IEC 25010	Drop (R1)

throughput requirements imposed by operational deadlines.

5. Resource Efficiency (RE)

Resource efficiency refers to the ability of an anomaly detector to operate within the computational, memory, and energy constraints of its target hardware, which in CPS/ICS environments is often constrained. It reflects the practicality of deploying the detector continuously on the intended platform, rather than relying on high-performance computing resources.

6. Adversarial Robustness (AR)

Adversarial robustness refers to the ability of anomaly detector to withstand deliberate attempts by an adversary to evade or compromise its operation. These attacks include evasion, data poisoning, and false-data-injection attacks. This criterion reflects the detector's ability to maintain reliable detection performance even when an attacker actively manipulates the input data or operating environment to bypass detection.

7. Data Confidentiality (CONF)

Data confidentiality refers to the ability of an anomaly detector to protect sensitive operational and process data from unauthorized disclosure throughout data collection, processing, storage, and transmission. It also encompasses mechanisms that minimize the exposure of sensitive raw data during detector training and operation.

8. Explainability (EXP)

Explainability refers to the ability of an anomaly detector to produce outputs that are understandable and traceable. It reflects whether operators can understand the reasons for a detection and whether the decision process is sufficiently documented for subsequent review and analysis. This enables alarms to be interpretable, actionable, and trustworthy rather than opaque.

9. Verification (VER)

Verification refers to whether there is credible evidence that the detector works as intended beyond a single benchmark result. It captures the depth of validation supporting a method, such as testing across multiple settings, ablation studies, runtime monitoring, or certification-oriented evidence, reflecting how confidently the detector could be relied upon in a real safety-critical deployment

6 Assessment of Approaches against Deployability Criteria

This section assesses how the reviewed studies address the deployability criteria by examining which criteria each paper explicitly treats as a design goal or evaluates experimentally. Each work is assigned one of three labels for each criterion. A criterion is marked as *Fully considered* when it is explicitly targeted and design choices are made to support it, or it is evaluated using appropriate metrics. A criterion is marked as *Partially considered* when it is mentioned or implied through design choices but is not explicitly targeted or evaluated. A criterion is marked as *Not considered* when the paper provides no discussion, design choice, or evaluation related to it.

Table 9 summarizes the assessment of how each reviewed work considers the deployability criteria. The labels indicate evidence provided or stated in the paper, not assumptions about the approach's full capabilities. If a criterion is not described or evidenced in the text, it is marked as 'not considered' here, without implying that the method itself cannot support it.

The table highlights that the criteria coverage is uneven across the literature. While most works provide strong evidence for detection accuracy, other criteria that are critical for deployability are less consistently considered. Some mechanisms are reported as noted below to support more than one criterion, especially operational resilience and adversarial robustness. However, they are credited where the paper explicitly defines and evaluates the relevant threat or failure mode.

Zhu et al. [62] is labeled fully for accuracy, reporting high F1 and recall on the SWaT and WADI benchmarks. Operational resilience is fully considered, as FFT-based sensor selection and variation-driven predicates smooth noise to stabilize detection. Adaptability is partially considered, as unsupervised invariant mining supports behavioral variation, but drift and regime changes are not explicitly targeted or evaluated. Real-time responsiveness is fully considered, with very low per-sample latency in an online detector. Resource efficiency is partially considered, since the lightweight online design implies low cost, but resource usage is not mea-

Table 8 Deployability criteria for anomaly detectors

Criteria	Acronym	Short description
Detection Accuracy	ACC	Ability to correctly detect anomalous behavior while controlling false positives and false negatives.
Operational Resilience	OR	Ability to maintain reliable operation under degraded conditions (e.g., missing/noisy data, sensor faults, unexpected failures) without unsafe behavior.
Adaptability	AD	Ability to maintain dependable detection when data distributions or operating conditions change over time.
Real-Time Responsiveness	RTR	Ability to produce detections within strict latency/throughput constraints so that mitigation actions remain timely.
Resource Efficiency	RE	Ability to meet detection objectives within energy/compute/resource budgets, supporting continuous operation on constrained platforms.
Adversarial Robustness	AR	Ability to sustain dependable detection under adversarial manipulation and other security-facing threats.
Data Confidentiality	CONF	Ability to protect the confidentiality of operational/process data during collection, processing, storage, and transmission, and to support privacy-preserving learning when required.
Explainability	EXP	Ability to provide interpretable, auditable detections that support understanding, operator diagnosis, and traceability of anomaly decisions.
Verification	VER	Ability to provide assurance evidence beyond a single test split (verification and validation, cross-testbed evaluation, run-time monitoring, conformance).

sured. Adversarial robustness is fully considered, as the method is explicitly motivated by and evaluated on stealthy attacks. Data confidentiality is not considered. Explainability is partially considered, since the mined invariant rules are interpretable and reflect physical relationships. Verification is fully considered, as the rule set is validated across two testbeds.

Zhang et al. [60] is labeled fully for accuracy, reporting multiple metrics on two datasets. Operational resilience is fully considered, since the PLS-TWE design and KS-entropy

monitoring improve reliability under fluctuating data. Adaptability is not considered, as only a generalization of the least-squares model is discussed, and drift is not addressed. Real-time responsiveness is fully considered, with training and detection times reported as a goal. Resource efficiency is partially considered, since optimised least-squares and entropy modules report low time cost, yet resource usage is not measured. Adversarial robustness is not considered. Data confidentiality is not considered. Explainability is partially considered, through an anomaly-localization step that identifies likely source components. Verification is fully considered with reported cross-dataset validation.

Jin et al. [27] is labeled fully for accuracy, reporting high detection rates across multiple IEEE bus systems. Operational resilience is fully considered, validated under different noise levels, measurement redundancies, PMU placements, and network sizes. Adaptability is not considered. Real-time responsiveness is fully considered, since the detection stage adds negligible computation over state estimation. Resource efficiency is partially considered, as execution times are reported and scalability to large networks is demonstrated. Adversarial robustness is not considered. Data confidentiality is not considered. Explainability is fully considered, as discriminating anomaly types and flagging the implicated loads or branches is a specifically evaluated output. Verification is fully considered with validation across multiple IEEE bus systems.

Hosseinzadehtaher et al. [19] is labeled partially for accuracy, as detection is demonstrated through case studies without quantitative metrics. Operational resilience is fully considered, as the CMV and USTDF design uses an error band around the forecast to handle normal variability. Adaptability is partially considered through a regression on historical load and temperature, but drift handling is not explicitly evaluated. Real-time responsiveness is fully considered, as AMI readings are continuously compared with ultra-short-term forecasts in an online monitor. Resource efficiency is not considered, as computation overheads are not discussed. Adversarial robustness is fully considered, given that the attack model explicitly covers stealthy device manipulation. Data confidentiality is not considered. Explainability is fully considered, as the CMV and decision-making module are designed to identify suspicious clusters and devices. Verification is partially considered, as analyses are verified through several case studies, though on a single simulated system.

Birihanu and Lendák [7] is labeled fully for accuracy, reporting metrics on SWaT and HAI. Operational resilience is not considered, with no design or evaluation for degraded conditions. Adaptability is partially considered, since correlations can be re-estimated, but drift is not targeted. Real-time responsiveness is partially considered, with lightweight correlation computation noted but latency not measured.

Resource efficiency is fully considered, as execution time and memory usage are explicitly measured across datasets to establish feasibility in resource-constrained environments. Adversarial robustness is not considered. Data confidentiality is not considered. Explainability is fully considered, as interpretability is the central goal: correlation-based attributions indicate which related parameters drive an alarm. Verification is fully considered, through cross-testbed evaluation on SWaT, HIL-HAI, and IoT-Modbus.

Yang et al. [56] is labeled fully for accuracy, reporting detailed metrics on the CPAD dataset. Operational resilience is not considered, as no mechanism for degraded or faulty conditions is provided. Adaptability is not considered, since distribution shift and evolving normal behaviour are not addressed. Real-time responsiveness is fully considered, since a maximum detection time per packet is set and the average is shown to stay below it. Resource efficiency is partially considered, as protocol-based packet filtering is explicitly used to improve detection efficiency, though resource budgets are not evaluated. Adversarial robustness is partially considered, as the FDIA threat model assumes a knowledgeable attacker and detection is evaluated under these manipulations, though detector-targeted evasion is not tested. Data confidentiality is not considered, since payload inspection is assumed. Explainability is not considered. Verification is partially considered, using a dedicated parameter-selection set with a stability re-check on the test set, but on a single testbed.

Mokhtari et al. [36] is labeled fully for accuracy, reporting accuracy, precision, recall, F1, and ROC/AUC for several supervised models on a hardware-in-the-loop attack dataset, with Random Forest performing best. Operational resilience is not considered, as the work is framed as fault/attack detection without evaluation under noisy, missing, or degraded data. Adaptability is not considered. Real-time responsiveness is partially considered, as fast prediction is stated as a requirement for real-time use and computation times are reported, though latency limits are not analysed. Resource efficiency is partially considered, since feature selection is explicitly motivated by reduced computation time. Adversarial robustness is fully considered, as detecting stealthy attacks that imitate normal behaviour is the explicit target and is evaluated on the HIL testbed. Data confidentiality is not considered. Explainability is not considered. Verification is not considered, as evaluation is limited to a single dataset and split.

Liu et al. [33] is labeled fully for accuracy, evaluated on three ICS datasets with reported metrics. Operational resilience is not considered, with no failure-condition design or evaluation. Adaptability is not considered, as drift appears only in related work. Real-time responsiveness is not considered, with no latency or throughput discussion. Resource efficiency is not considered. Adversarial robustness is not

considered, as the OTS attack dataset supports detection evaluation, not adversarial manipulations. Data confidentiality is not considered, assuming full data access. Explainability is not considered. Verification is fully considered, via extensive ablation experiments and validation on two public and one self-built dataset.

Yang et al. [57] is labeled fully for accuracy, reporting detailed metrics on SWaT, WADI, and HAI where ADT outperforms baselines. Operational resilience is fully considered, with explicit robustness studies under delayed, partial, and noisy feedback. Adaptability is fully considered, as dynamic threshold selection learns effective policies with minimal labels. Real-time responsiveness is fully considered, presenting a lightweight dynamic-thresholding controller for real-time detection. Resource efficiency is fully considered, as the controller is presented as lightweight to support real-time operation. Adversarial robustness is not considered. Data confidentiality is not considered. Explainability is not considered. Verification is fully considered, through cross-testbed evaluation on SWaT, WADI, and HAI together with robustness studies.

Zare et al. [59] is labeled fully for accuracy, reporting per-attack recall gains over two network anomaly detectors on SWaT traffic. Operational resilience is not considered, with no evaluation under degraded or faulty conditions. Adaptability is partially considered, as the LSTM-autoencoder sets a precision-recall-curve threshold but does not explicitly handle drift. Real-time responsiveness is fully considered, being built to process packet sequences online. Resource efficiency is not considered. Adversarial robustness is not considered, as attack detection on SWaT traffic is evaluated without a stealthy or detector-aware manipulation model. Data confidentiality is not considered, assuming full traffic access. Explainability is not considered. Verification is not considered, with single-testbed evaluation only.

Liao et al. [32] is labeled fully for accuracy, with reported evaluation metrics. Operational resilience is partially considered, via connection-stability scoring that addresses noise sensitivity. Adaptability is partially considered, as the dynamic graph updates per batch, though long-term drift is not targeted. Real-time responsiveness is partially considered, motivated by real-time monitoring but without latency measurement. Resource efficiency is partially considered, discussing VAE and LSTM efficiency but not quantifying it. Adversarial robustness and data confidentiality are not considered. Explainability is partially considered, through a dedicated anomaly-explanation analysis. Verification is fully considered, via ablation and hyperparameter experiments and validation on six datasets.

Aslam et al. [6] is labeled fully for accuracy, with the improved autoencoder (ImpAE) reporting full metrics on the SWaT benchmark. Operational resilience is not considered, as missing-value handling is data cleaning rather

Table 9 Assessment of the reviewed approaches against the deployability criteria

Research Work	Family	ACC	OR	AD	RTR	RE	AR	CONF	EXP	VER
Zhu et al. [62]	Mathematical	●	●	●	●	●	●	○	●	●
Zhang et al. [60]	Mathematical	●	●	○	●	●	○	○	●	●
Jin et al. [27]	Mathematical	●	●	○	●	●	○	○	●	●
Hosseinzadehtaher et al. [19]	Mathematical	●	●	●	●	○	●	○	●	●
Birihanu and Lendák [7]	Machine Learning	●	○	●	●	●	○	○	●	●
Mokhtari et al. [36]	Machine Learning	●	○	○	●	●	●	○	○	○
Yang et al. [56]	Deep Learning	●	○	○	●	●	●	○	○	●
Liu et al. [33]	Deep Learning	●	○	○	○	○	○	○	○	●
Yang et al. [57]	Deep Learning	●	●	●	●	●	○	○	○	●
Zare et al. [59]	Deep Learning	●	○	●	●	○	○	○	○	○
Liao et al. [32]	Deep Learning	●	●	●	●	●	○	○	●	●
Aslam et al. [6]	Deep Learning	●	○	○	○	●	○	○	●	○
Tang et al. [53]	Deep Learning	●	○	○	○	●	○	○	●	●
Macas et al. [34]	Deep Learning	●	●	●	○	●	○	○	○	○
Kea et al. [28]	Deep Learning	●	●	●	●	●	○	●	○	●
Sahin et al. [47]	Deep Learning	●	●	○	○	●	○	○	○	○
Do Thu Ha et al. [14]	Deep Learning	●	○	○	●	●	○	○	●	○
Zhou et al. [61]	Deep Learning	●	○	●	●	●	○	○	○	○
Huong et al. [22]	Deep Learning	●	●	●	●	●	○	●	○	●
Jeffrey et al. [26]	Hybrid	●	●	●	●	●	○	○	●	●
Hu et al. [20]	Hybrid	●	●	●	●	●	○	○	○	●
Hao et al. [15]	Hybrid	●	●	●	●	●	○	○	○	●
Asefi et al. [5]	Hybrid	●	●	●	●	●	●	○	●	●
Wu et al. [55]	Hybrid	●	○	○	○	●	○	○	●	●
Raman and Mathur [45]	Hybrid	●	●	○	●	●	●	○	●	●
Feng and Tian [11]	Hybrid	●	●	○	●	●	○	○	○	●
Fully (●)		25	9	3	13	6	6	2	6	15
Partially (●)		1	7	11	7	17	1	0	8	4
Not (○)		0	10	12	6	3	19	24	12	7

● Fully considered: explicitly targeted as a design goal and supported by design choices, or evaluated with appropriate metrics.

● Partially considered: mentioned or implied through a design choice but not explicitly targeted or evaluated.

○ Not considered: no discussion, design goal, or evaluation related to the criterion

than resilience under degraded conditions. Adaptability is not considered, with no scheme for evolving normal behaviour. Real-time responsiveness is not considered, as real-time detection appears only in related work. Resource efficiency is partially considered, targeting reduced computational complexity but not quantifying it. Adversarial robustness is not considered, as adversarial aspects appear only in related work. Data confidentiality is not considered. Explainability is partially considered, providing sensor- and actuator-level

indications of flagged points. Verification is not considered, as it uses standard splits on SWaT only.

Tang et al. [53] is labeled fully for accuracy, with the GRU-based detector reporting full metrics on SWaT and WADI. Operational resilience is not considered. Adaptability is not considered. Real-time responsiveness is not considered. Resource efficiency is partially considered, noting a shallow-parameter network without benchmarking it. Adversarial robustness is not considered. Data confidentiality is not considered. Explainability is fully considered, since inter-

pretability is a stated design goal producing interpretable outputs. Verification is fully considered, via cross-testbed evaluation on SWaT and WADI.

Macas et al. [34] is labeled fully for accuracy, with the attention-based deep generative model reporting the full metric suite on SWaT. Operational resilience is partially considered, testing detection under RBF-kernel perturbation and noise, though not framed as failures. Adaptability is partially considered, through a state-based dynamic threshold that adapts to the estimated state of task execution. Real-time responsiveness is not considered. Resource efficiency is partially considered, mentioning architecture efficiency without measuring it. Adversarial robustness is not considered. Data confidentiality is not considered. Explainability is not considered. Verification is not considered, as evaluation is restricted to a single testbed with standard splits.

Kea et al. [28] is labeled fully for accuracy, with the evaluation of FLAE showing improved detection. Operational resilience is partially considered, motivated by heterogeneous IoT power data, though failures and noise are not tested. Adaptability is partially considered via federated retraining on updated local data, though concept drift is not studied. Real-time responsiveness is fully considered, as federated learning is argued to cut response-time delay relative to centralized deep learning. Resource efficiency is fully considered, since federated learning reduces central data transfer and computation costs. Adversarial robustness is not considered. Data confidentiality is fully considered, as privacy is the main motivation of the work. Explainability is not considered. Verification is fully considered, via an ablation study across configurations.

Sahin et al. [47] is labeled fully for accuracy, with the CNN-LSTM framework reporting precision, recall, and accuracy on water-distribution data. Operational resilience is partially considered, handling missing data via linear interpolation. Adaptability is not considered. Real-time responsiveness is not considered. Resource efficiency is partially considered, mentioning model efficiency without measuring it. Adversarial robustness and data confidentiality are not considered. Explainability is not considered. Verification is not considered.

Do Thu Ha et al. [14] is labeled fully for accuracy, reporting precision, recall, F1, and ROC on the Gas Pipeline (Morris) dataset. Operational resilience is not considered, with no evaluation under degraded or noisy conditions. Adaptability is not considered. Real-time responsiveness is partially considered, aiming at timely operator response without benchmarking latency. Resource efficiency is partially considered, noting the efficiency of the LSTM-autoencoder/OCSVM design without measuring it. Adversarial robustness is not considered. Data confidentiality is not considered. Explainability is fully considered, as gradient-based SHAP attributions identify the measurements driving each detection.

Verification is not considered, with single-dataset evaluation and no assurance evidence.

Zhou et al. [61] is labeled fully for accuracy, reporting performance metrics for detection with few labels. Operational resilience is not considered. Adaptability is partially considered, as few-shot learning generalizes to novel classes, though concept drift is not addressed. Real-time responsiveness is partially considered, as real-time analysis is stated as a design intent but no mechanism or measurement is provided. Resource efficiency is partially considered, focusing on sample efficiency rather than compute efficiency. Adversarial robustness, data confidentiality, explainability and verification are not considered.

Huong et al. [22] is labeled fully for accuracy, reporting detection metrics on ICS datasets. Operational resilience is fully considered, explicitly targeting contextual noise, missing data, and concept drift in edge IIoT. Adaptability is partially considered, via an adaptive kernel-quantile-estimator threshold and a no-retraining argument, though drift handling is not evaluated. Real-time responsiveness is fully considered, moving detection to edge devices to avoid cloud delay. Resource efficiency is fully considered, showing that federated learning saves bandwidth and reduces transmission time. Adversarial robustness is not considered. Data confidentiality is fully considered, as federated learning keeps data local as a central design goal. Explainability is not considered. Verification is fully considered, with evaluation across multiple datasets and a multi-node edge deployment.

Jeffrey et al. [26] is labeled fully for accuracy, with the hybrid threshold-and-ML detector reporting accuracy and false-alarm rate. Operational resilience is partially considered, as environmental fluctuations are explicitly accounted for in the training-range design. Adaptability is partially considered, as a stochastic threshold accommodates communication variability, though drift is not targeted. Real-time responsiveness is partially considered, since IT-network threshold checks imply timeliness but latency is not measured. Resource efficiency is partially considered, discussing reduced IDS computational complexity. Adversarial robustness is not considered. Data confidentiality is not considered. Explainability is partially considered, since any-vote alarms are tied to physical-tolerance thresholds, giving operators interpretable signals. Verification is partially considered, through ten-times-repeated ten-fold cross-validation and an independent final validation stage, though on a single testbed.

Hu et al. [20] is labeled fully for accuracy, reporting detection accuracy and state-estimation RMSE on IEEE 14-, 30-, and 118-bus systems. Operational resilience is fully considered, explicitly handling measurement noise, bad data, and sudden load changes via forecast-assisted state estimation. Adaptability is fully considered, as adaptive adjustment of the smoothing parameters is the core design mechanism. Real-time responsiveness is fully considered, as maintain-

ing real-time state estimation is an explicit design constraint of the compensation scheme. Resource efficiency is partially considered, as the EKF is computationally light but complexity is not measured. Adversarial robustness is fully considered, with an explicit false-data-injection and sudden-load-change threat model evaluated against attacks. Data confidentiality is not considered. Explainability is partially considered, localising and correcting compromised measurements. Verification is fully considered, validated across IEEE 14-, 30-, and 118-bus systems.

Hao et al. [15] is labeled fully for accuracy, with the hybrid statistical-ML detector reporting metrics on ICS traffic. Operational resilience is partially considered, modelling the regular operating schedule, though failures are not tested. Adaptability is partially considered, noting the need for retraining for transferability but offering no online scheme. Real-time responsiveness is fully considered, as real-time detection is the central goal through a scalable, efficient pipeline. Resource efficiency is fully considered, being explicitly scalable and efficient with low time cost. Adversarial robustness is not considered. Data confidentiality is not considered. Explainability is not considered. Verification is fully considered, validated on a realistic testbed spanning three ICPSs (power generation, gas pipeline, and urban railway) across a range of scenarios.

Asefi et al. [5] is labeled fully for accuracy, combining model-based and data-driven components with full metrics on the IEEE 14-bus system. Operational resilience is fully considered, evaluated under measurement noise, bad data, and load/topology changes. Adaptability is fully considered, as the ML component is explicitly trained to be topology-independent, eliminating retraining after network reconfiguration. Real-time responsiveness is fully considered, with millisecond real-time prediction times measured and complexity analysed. Resource efficiency is fully considered, through time-complexity analysis and measured runtime on modest hardware. Adversarial robustness is fully considered, explicitly targeting false-data-injection attacks on state estimation. Data confidentiality is not considered. Explainability is partially considered, separating anomaly types (bad data, attack, change) interpretably. Verification is partially considered, through cross-condition validation on the IEEE 14-bus system.

Wu et al. [55] is labeled fully for accuracy, with PGRGAT reporting precision, recall, F1, and AUPRC on SWaT and WADI and outperforming data-driven baselines. Operational resilience is not considered. Adaptability is not considered. Real-time responsiveness is not considered. Resource efficiency is partially considered, discussing model complexity without benchmarking it. Adversarial robustness is not considered. Data confidentiality is not considered. Explainability is fully considered, as the physics-informed graph structure improves interpretability—a stated contribution. Verification

is fully considered, via ablation, statistical significance testing, and evaluation on SWaT and WADI.

Raman and Mathur [45] is labeled fully for accuracy, with the physics-based, data-driven PbNN reporting detection rate and false-alarm rate. Operational resilience is partially considered, noting short-term deviations from sensor noise and irregularity. Adaptability is not considered. Real-time responsiveness is fully considered, implemented and run in an operational secure water-treatment plant with live attacks. Resource efficiency is partially considered, motivated by a lightweight design but not benchmarked. Adversarial robustness is fully considered, launching and evaluating real-time stealthy and coordinated attacks on the live plant. Data confidentiality is not considered. Explainability is partially considered, as residue-based estimates indicate anomalous process variables. Verification is fully considered, validated on an operational plant with live attacks and a multi-plant cross-check.

Feng and Tian [11] is labeled fully for accuracy, with NSIBF (neural system identification plus Bayesian filtering) reporting metrics on three datasets. Operational resilience is fully considered, as Bayesian filtering explicitly handles measurement noise and uncertainty. Adaptability is not considered. Real-time responsiveness is partially considered, as sequential filtering suits streaming though latency is not measured. Resource efficiency is partially considered, noting a lightweight state-space model without benchmarking it. Adversarial robustness is not considered. Data confidentiality is not considered. Explainability is not considered. Verification is fully considered, via cross-dataset validation on SWaT, WADI, and PUMP.

7 Discussion

Recall that the screening process resulted in 26 papers, including 4 mathematical, 2 machine-learning, 13 deep-learning, and 7 hybrid studies. The corpus is clearly dominated by deep learning, which reflects its dominance in the current literature. In parallel, hybrid approaches also account for a substantial share of the reviewed work. This is notable as the assessment suggests that hybrid detectors are among the most deployment-oriented overall. In terms of evaluation data, the studies rely heavily on a small number of recurring benchmark datasets, such as SWaT and WADI. This concentration indicates that much of the evidence base is built around a limited number of well-established test environments. As a consequence, the reported accuracy is subject to several evaluation pitfalls, including point-adjusted scoring that can inflate F1, optimistic AUROC values under the severe class imbalance typical of CPS/ICS datasets, and dataset-specific overfitting resulting from label noise and repeated tuning on the same public test splits.

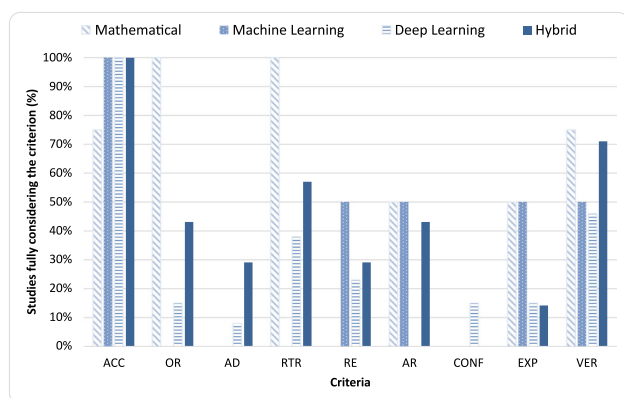


Fig. 3 Criteria coverage by method family: share of studies in each method family that fully consider each criterion

The assessment in Table 9 revealed the differences in how the reviewed detectors address deployability. Detection accuracy is the criterion that is almost universally fully considered, whereas consideration drops substantially for the remaining criteria. However, even for accuracy, the reported scores are often measured using evaluation protocols that can overstate real-world performance. For example, a detector may appear high-performing on the basis of high benchmark scores despite producing frequent false alarms.

Real-time responsiveness and verification receive substantial attention, although verification evidence takes the form of cross-testbed validation rather than formal assurance artifacts. Operational resilience is polarized, as several works explicitly design and evaluate for degraded conditions, while a comparable number do not consider them at all. Resource efficiency is mostly implied through lightweight designs but rarely measured. Adversarial robustness, adaptability, and data confidentiality are the weakest covered criteria. The results show that the literature is still organized around benchmark performance first, with deployability concerns treated as secondary.

Mathematical detectors are the most consistent in operational resilience and real-time responsiveness, largely because their formulations are lightweight and explicitly tied to noise handling and timing. Deep-learning detectors broaden the range of criteria considered, but most individual studies still mostly address accuracy and only a few additional properties. Machine-learning detectors also mainly reinforce the strong focus on accuracy. Hybrid detectors are broadly balanced across the criteria, which suggests that combining model-based and learned components can better support deployability.

The family coverage shown in Figure 3 reinforces this interpretation. All four families fully address accuracy, but they diverge on the remaining criteria.

Some criterion pairings also recur in ways that are not accidental, as illustrated in Figure 4. In particular, real-time

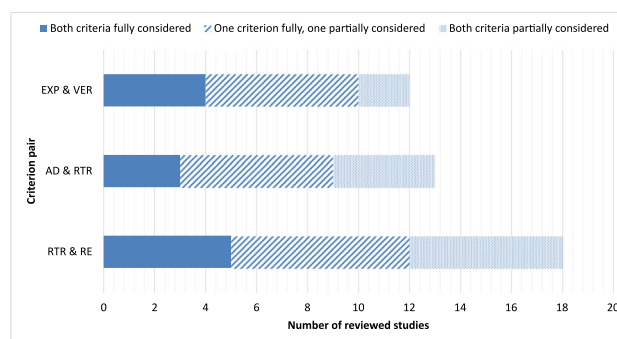


Fig. 4 Co-occurrence of criterion pairs

responsiveness and resource efficiency often appear together, as in many cases they are driven by the same deployment constraint. For example, methods intended for online or edge-oriented operation, usually must satisfy both latency and computational resource constraints. Adaptability and real-time responsiveness are likewise closely connected, since detectors that adapt to changing conditions are often designed for continuous online operation. Explainability and verification also form a meaningful pairing, as both contribute to confidence, auditability, and stronger assurance arguments.

The opposite pattern can also be observed for data confidentiality and explainability, as no reviewed study considers both criteria. The few studies that target confidentiality do so through federated architectures that keep data local, without providing evidence for considering explainability. This suggests a trade-off between privacy-preserving learning and interpretable detection.

It is worth noting that some criteria are more important in certain sectors than in others. For instance adversarial robustness and operational resilience are especially important in water and power environments, where malicious manipulation and degraded measurements can directly affect system operation. Real-time responsiveness becomes more critical in domains with strict timing requirements, such as power grid protection, where delayed anomaly detection may prevent timely intervention. In contrast, data confidentiality is more relevant in settings such as geographically distributed critical infrastructure, where sensitive operational data cannot be freely centralized or shared. Therefore, the nine defined criteria are intended to provide a common assessment framework, with their relative importance depending on the deployment context.

To summarize, the corpus suggests that anomaly-detection research focuses on detection performance and latency, while adversarial robustness, adaptability, and confidentiality remain underdeveloped. The most deployment-oriented designs appear in the hybrid and mathematical families. Evaluating anomaly detectors against the deployability criteria and reporting evidence for these criteria more systematically

should guide the development of more deployable anomaly detectors.

8 Threats to Validity

The review should also be read with its scope constraints in mind. The review is deliberately limited to implementation-based anomaly detection for industrial CPS/ICS, rather than CPS or safety-critical systems in general. This was a deliberate choice, as our aim was to establish a focused evidence base in which deployability could be examined, rather than to conduct an exhaustive review of anomaly-detection research. As a consequence, the corpus exhibits some concentration in datasets and application domains, clustering around a small number of recurring testbeds. This reflects the deliberately narrow scope of the review rather than a sampling error, but it does limit the generalizability of the findings.

The search strategy is a second source of limitation. The Boolean query was applied only to paper titles to improve precision, thereby retaining studies in which anomaly detection is the primary contribution rather than those that merely mention a detector as one component of a larger CPS/ICS solution. The trade-off introduced is reduced recall, since relevant studies whose titles do not explicitly mention the topic may have been missed. Similarly, the query focused on anomaly detection and CPS/ICS domain terms while intentionally omitting method-family terms to avoid biasing the corpus toward particular approaches.

Finally, the three-level labeling involves interpretive judgment. To reduce subjectivity, each study was labeled independently by two assessors, with disagreements resolved through discussion, and inter-rater agreement is reported.

9 Conclusion and Future Work

This survey examined how well recent anomaly detection methods for industrial CPS/ICS are evaluated against the demands of real-world deployment. Its primary contribution is the definition of nine deployability criteria, derived from established safety, security, and AI-governance standards and frameworks, together with an evidence-based assessment of 26 studies against these criteria. The assessment reveals a clear imbalance across the coverage of the criteria. Detection accuracy is the only criterion that is almost universally considered fully, whereas other properties that affect operational trust are addressed far less consistently. Current evaluation practice remains largely focused on benchmark performance, leaving a gap between reported performance and deployment readiness.

Beyond the individual results, these findings point to a broader implication. Without a unified deployment frame-

work, it is difficult for practitioners to determine which detectors are the most deployment-ready and for researchers to identify where progress is most needed. The proposed criteria and the resulting assessment offer such a framework, making these gaps explicit and comparable across different methods. Importantly, the criteria serve a dual role. First, they can support the evaluation of detector deployability. Second, they can serve as design guidelines by specifying the properties that detectors should possess.

These results also indicate where future work is most valuable. Future evaluation should move beyond accuracy by developing and evaluating detectors against the deployability criteria that are currently under-supported. For example, directions could include drift injection and adversary testing, measurement of both latency and resource usage, cross-testbed and cross-domain transfer, calibrated-confidence or abstention mechanisms, and operator-in-the-loop assessment. Because the relative importance of the criteria varies across application setting, such evaluations should be adapted to the target domain rather than applying a uniform approach. Finally, deployability should be treated as a lifecycle concern, supported by appropriate measures throughout all phases, so that these properties are planned and designed rather than added as an afterthought. By reframing both design and evaluation around deployability and identifying the current gaps, this survey aims to provide guidelines for future research on anomaly detection with more comprehensive coverage of deployability criteria.

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Data Availability No datasets were generated or analysed during the current study.

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